



Kubernetes Error 4: Node NotReady

- A Kubernetes node enters the *NotReady* state when it cannot communicate properly with the control plane or has resource/system issues •

Common reasons include:

- Disk Pressure
- Memory Pressure
- PID Pressure (*Process IDs are exhausted*)
- Network connectivity issues
- Kubelet service stopped or unhealthy
- Container runtime (**Docker/containerd**) failure

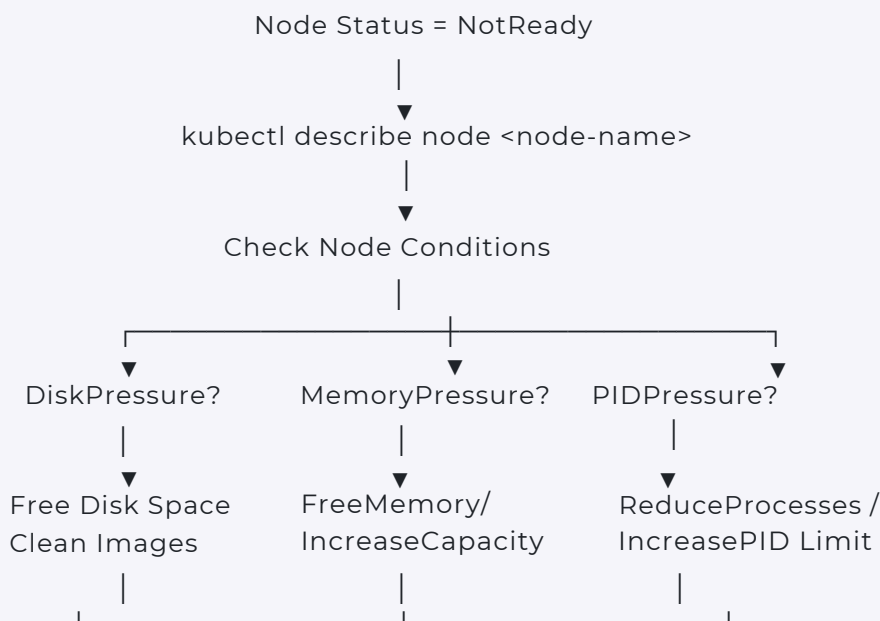
✓ Check the node status

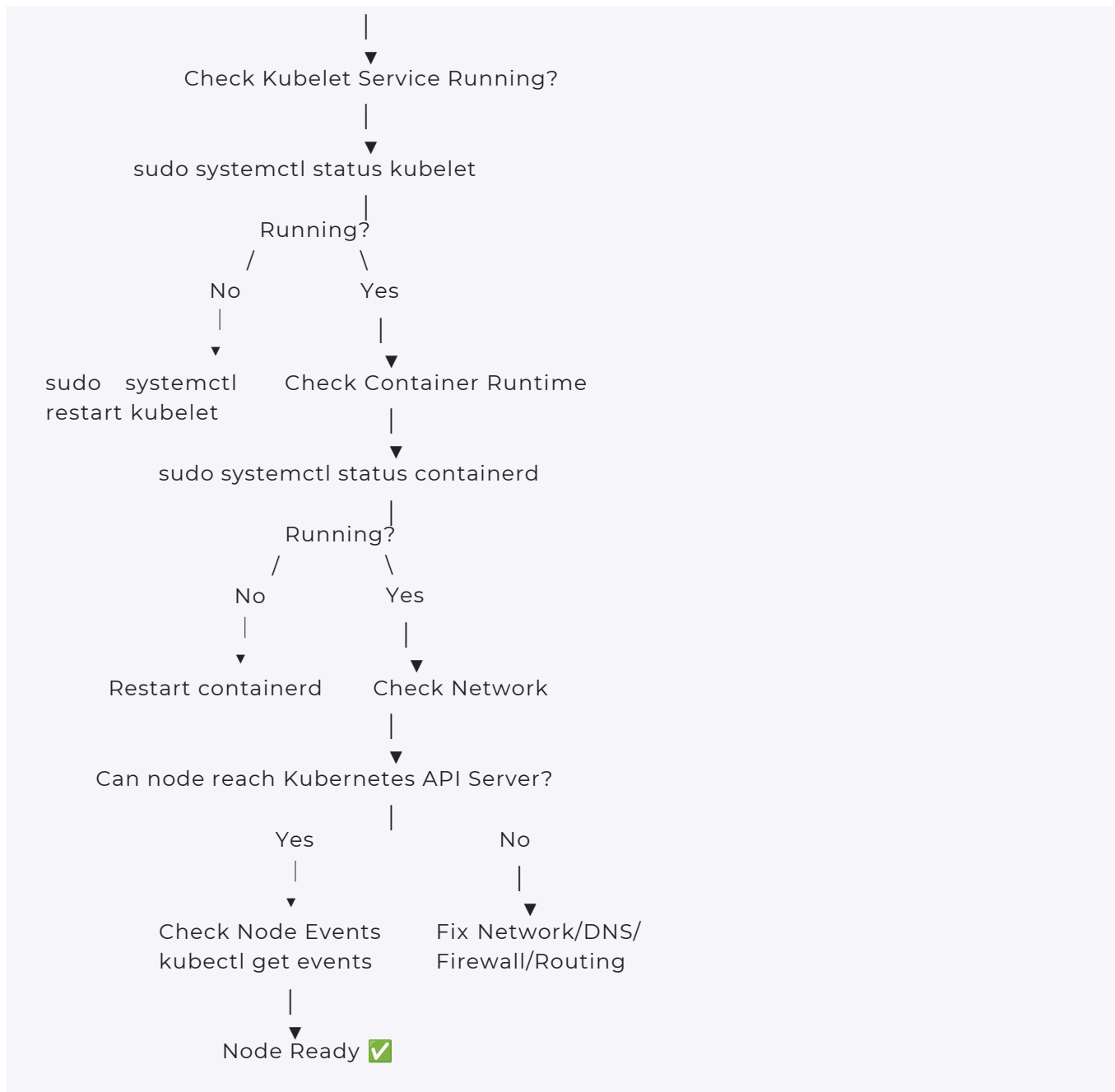
- ***kubectl get nodes***

NAME	STATUS	ROLES	AGE	VERSION
worker1	NotReady	<none>	15d	v1.31.0



Kubernetes Node NotReady Troubleshooting Flow





Kubernetes Node NotReady — Common Reasons & Solutions

 Cause	 Description	 Solution	 Useful Commands
 Disk Pressure	Node is running out of disk space	Free disk space, remove unused images/logs, or expand the disk	<code>kubectl describe node <node-name></code> <code>df -h</code>
 Memory Pressure	Node has insufficient available memory	Stop unnecessary processes, optimize	<code>kubectl describe node <node-name></code> <code>free -h</code>

 Cause	 Description	 Solution	 Useful Commands
		workloads, or add memory	
 PID Pressure	Node has exhausted available process IDs	Reduce running processes or increase the PID limit	<code>kubectl describe node <node-name></code>
 Network Issue	Node cannot communicate with the control plane	Verify network connectivity, security groups, firewall, and routes	<code>ping / curl / kubectl describe node</code>
 Kubelet Not Running	Kubelet service has stopped or failed	Restart kubelet and inspect its logs	<code>systemctl status kubelet</code> <code>journalctl -u kubelet</code>
 Container Runtime Failure	Docker/containerd is not running	Restart the container runtime and review logs	<code>systemctl status containerd</code>
 Certificate Expired	Kubelet certificate has expired	Renew the certificate and restart kubelet	<code>kubeadm certs renew (kubeadm clusters)</code>
 Cloud Provider Issue	EC2 instance or cloud networking problem	Verify the VM is healthy and networking is configured correctly	Check cloud console and instance status
 Node Shutdown/Reboot	Node has restarted or is powered off	Bring the node back online and verify services	<code>kubectl get nodes</code>
 High CPU/Memory Usage	Node is overloaded	Reduce workload, add nodes, or enable Cluster Autoscaler	<code>kubectl top node</code>

Useful Troubleshooting Commands

 Command	 Purpose
<code>kubectl get nodes</code>	Check node status
<code>kubectl describe node <node-name></code>	View node conditions and events
<code>kubectl top node</code>	Check CPU and memory usage
<code>systemctl status kubelet</code>	Verify kubelet service
<code>journalctl -u kubelet -f</code>	View kubelet logs
<code>systemctl status containerd</code>	Verify container runtime
<code>df -h</code>	Check disk usage
<code>free -h</code>	Check memory usage

Interview Answer

Q: How do you troubleshoot a Node Not Ready issue ?

- Run **kubectl get nodes** to identify the affected node.
- Use **kubectl describe node <** to inspect node conditions and events.
- Check for disk, memory, or PID pressure (Process IDs are exhausted).
- Verify that the **kubelet** and **container runtime** are running.
- Check network connectivity between the node and the control plane.
- Review kubelet logs and resolve any underlying infrastructure or resource issues.

Interview One-Liner

- A Node NotReady status means the control plane considers the **node unhealthy** or **unreachable**.
- Common causes include disk pressure, memory pressure, PID exhaustion, kubelet or container runtime failures, network connectivity issues, certificate problems, or cloud infrastructure issues.
- The first troubleshooting step is to inspect the node using **kubectl describe node <node-name>**.